

Reconstructing Human Object Interactions from Monocular Videos

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Abstract

Reconstructing 3D human-object interactions (HOI) from video is challenging due to depth ambiguity, occlusions, and inconsistencies between human and object pose estimation frameworks. We present a hybrid pipeline that reconstructs temporally coherent 3D human and object poses by combining monocular human motion recovery with multi-view object tracking. Specifically, human pose is estimated from a single RGB view using GVHMR, while object pose is robustly tracked across multiple RGB-D views using FoundationPose. To bridge these distinct representations, we apply a coordinate-frame unification procedure, a depth-scale correction derived from per-frame BEHAVE intrinsics, and a SMPL-X to SMPL pelvis offset. The resulting human and object estimates are transformed into a shared coordinate frame and jointly refined through optimization with fidelity, contact, penetration, and temporal smoothness objectives.

We evaluate our approach on interaction sequences from the BEHAVE dataset. Our results show that accurate, physically plausible human-object reconstruction is achievable by integrating complementary estimation models within a unified optimization framework. Furthermore, we find that occlusion remains the primary factor limiting reconstruction accuracy, highlighting the need for stronger object reasoning and interaction-aware modeling in future HOI systems.

Supplementary visualization videos are included in our submission materials, and the source code is publicly available at <https://github.com/Marco-Weder/Reconstructing-Human-Object-Interactions-from-Monocular-Videos>.

1. Introduction

Reconstructing how humans interact with objects is a core challenge in computer vision, especially for applications like robotics and augmented reality. The objective is to capture both the human body and the manipulated object

within a shared 3D coordinate space, ensuring their spatial and temporal interactions remain physically plausible without artifacts like floating or mesh clipping.

Despite significant progress in human motion recovery and object pose estimation in isolation, joint human-object interaction reconstruction remains challenging. Monocular human reconstruction suffers from depth ambiguity and occlusion, while object pose estimation becomes increasingly unreliable when large portions of an object are hidden by the interacting person. Furthermore, state-of-the-art human and object reconstruction systems are typically developed independently and operate in incompatible coordinate frames. Differences in camera calibration, body model representations, and world coordinate definitions make direct integration difficult, often resulting in spatially inconsistent reconstructions that fail to accurately capture physical interactions.

In this project, we present a pipeline that integrates GVHMR and FoundationPose within the BEHAVE benchmark framework. Human motion is reconstructed from a monocular camera using GVHMR, while object poses are obtained through a multi-camera FoundationPose setup to improve robustness under occlusion. We develop a coordinate-frame unification procedure that transforms human and object estimates into a common camera reference frame. To further improve consistency, we introduce depth-scale correction based on calibrated camera intrinsics and a pelvis-origin correction that aligns GVHMR’s SMPL-X output with the SMPL evaluation framework used by BEHAVE. The resulting reconstructions are further refined through joint optimization with contact, penetration, and temporal smoothness constraints.

We evaluate the proposed pipeline on interaction sequences from the BEHAVE dataset and analyze reconstruction performance across different interaction types. Our results demonstrate that integrating complementary human and object reconstruction systems can produce physically plausible human-object interactions without requiring depth supervision, while also highlighting the significant impact of occlusion on reconstruction accuracy.

Our contributions are summarized as follows. First, we demonstrate that monocular human reconstruction and multi-camera object tracking can be effectively integrated within a common coordinate framework for human-object interaction reconstruction. Second, we identify and address key sources of geometric inconsistency between the two systems through coordinate-frame and body-model alignment. Third, we provide an empirical evaluation on the BEHAVE dataset, including an analysis of how occlusion affects reconstruction performance.

2. Related Work

2.1. Human Motion Recovery

Recovering 3D human pose and shape from monocular images and videos has been extensively studied in computer vision. Early human mesh recovery methods, such as HMR [3], estimate the parameters of parametric body models from RGB observations, enabling reconstruction of full-body meshes from single images. More recent approaches incorporate temporal information from video sequences to improve motion consistency and reduce frame-to-frame jitter [4].

GVHMR (Gravity-View Human Motion Recovery) [6] extends this line of work by recovering world-grounded human motion from monocular video. Rather than predicting global motion step by step, GVHMR introduces a Gravity-View coordinate system that is defined by the gravity direction and camera view direction. This representation reduces ambiguity in world coordinate estimation and enables temporally coherent reconstruction of human motion in a gravity-aware reference frame. Due to its strong monocular reconstruction performance and availability of global human motion estimates, GVHMR serves as the human reconstruction component in our pipeline.

2.2. Object Pose Estimation and Tracking

Estimating the six-degree-of-freedom (6DoF) pose of objects from visual observations is a fundamental problem in computer vision and robotics. Traditional instance-level methods typically require object-specific training and struggle to generalize to unseen objects. Recent research has focused on developing methods that can estimate object poses for novel objects without retraining.

FoundationPose [7] is a unified framework for object pose estimation and tracking that supports both model-based and model-free settings. Given an RGB-D frame, a mask or bounding box of the target object, and either a CAD model or a small set of reference images, FoundationPose can estimate and track the pose of previously unseen objects through large-scale synthetic training and transformer-based feature learning. The method demonstrates strong robustness across diverse object categories and challenging

viewpoints. In our work, FoundationPose provides the initial object pose estimates used for human-object interaction reconstruction.

2.3. Human-Object Interaction Reconstruction and Evaluation

Understanding human-object interactions requires reasoning about both human motion and object motion within a shared spatial context. While substantial progress has been made in reconstructing humans and objects independently, jointly recovering their interactions remains challenging due to occlusions, contact constraints, and ambiguities in monocular observations.

The BEHAVE dataset [1] was introduced to facilitate research on 3D human-object interaction understanding. It contains multi-view RGB-D recordings of subjects interacting with everyday objects, together with fitted SMPL body models, object meshes, and contact annotations. The dataset provides one of the most comprehensive benchmarks for evaluating human-object interaction reconstruction and has been widely adopted in subsequent research. In this project, BEHAVE serves as both the source of interaction sequences and the evaluation benchmark.

Unlike prior work that focuses on either human reconstruction or object pose estimation independently, our project investigates how state-of-the-art human and object reconstruction systems can be integrated into a unified coordinate frame and refined through joint optimization. By combining GVHMR and FoundationPose within the BEHAVE evaluation framework, we study the challenges of monocular human-object interaction reconstruction, particularly under severe occlusion and complex physical interactions.

3. Method

3.1. FoundationPose Object Tracking

We use FoundationPose in its model-based configuration: the CAD mesh provided by BEHAVE is given to the network as the object prior, and per-frame RGB-D observations from the calibrated Kinect cameras drive registration and tracking. We rely on the depth channel because FoundationPose’s pose hypothesis sampling and refinement are inherently depth-based, and BEHAVE’s registered Kinect depth provides the required input directly. At initialization FoundationPose requires a 2D mask of the object in the first frame; we use the ground-truth object masks supplied by BEHAVE, since segmentation is not the focus of this work. In a deployment scenario without such annotations, an off-the-shelf zero-shot segmentation model such as CNOS [5] could be substituted without affecting the rest of the pipeline.

Depth-based tracking confidence. A common failure



Figure 1. **Object pose tracking under single-camera and multi-camera settings.** Multi-camera observations provide improved robustness to occlusion and viewpoint changes compared to a single-camera setup.

mode of feed-forward trackers is silent drift: pose estimates remain visually plausible long after the tracker has effectively lost the object. We therefore introduce a simple depth-consistency confidence score that is used throughout this subsection, both to detect single-camera failures and, later, to choose the active view in the multi-camera setting. Given the current pose $P \in \text{SE}(3)$, the camera intrinsics K , the observed depth image D , and the canonical mesh vertices $\{V_j\}_{j=1}^N$, we transform each vertex into the camera frame as $\tilde{V}_j = RV_j + t$, discard vertices that fall behind the camera ($\tilde{V}_j^{(z)} \leq 0.01 \text{ m}$) or project outside the image plane after applying K , and compare the remaining vertices’ rendered depth $\hat{z}_j = \tilde{V}_j^{(z)}$ against the observed depth $z_j = D[v_j, u_j]$ at the projected pixel. A vertex is counted as matched if the observed depth is valid ($z_j > 1 \text{ mm}$) and agrees with the predicted depth within a tolerance of $\tau = 5 \text{ cm}$:

$$\text{conf}(P) = \frac{|\{j : z_j > 0.001 \wedge |z_j - \hat{z}_j| < \tau\}|}{|\{j : \text{vertex } j \text{ is in front and in-image}\}|} \in [0, 1]. \quad (1)$$

Intuitively, this is the fraction of the visible mesh surface whose predicted geometry is supported by the observed depth map. It is cheap to compute, requires no learned rescorer, and degrades gracefully when the object becomes partially occluded.

Single-camera tracking. The most natural setting is to run FoundationPose on a single Kinect view, matching the monocular assumption of the rest of the pipeline. To keep the tracker robust under the partial occlusion that characterises BEHAVE, we use the confidence score above as a recovery signal: whenever $\text{conf}(P_t)$ falls below a re-initialization threshold, we re-register the pose from the ground-truth mask at that frame and resume tracking. This works well for short occlusions caused by an arm or hand passing in front of the object, where the tracker briefly loses confidence and then locks back on. Recovery is, however, fundamentally limited by what is visible in the frame: on sequences where the body completely occludes the object

for extended intervals – for example when an object is held against the torso or carried on the back – no amount of re-registration on the active view can recover information that is not in the image. This motivates extending the tracker to use the additional camera views available in BEHAVE.

Frame synchronization. A prerequisite for multi-camera tracking is that the four Kinect streams be aligned in time, since the cameras record with slightly different start times and occasional dropped frames. BEHAVE provides, for each camera, a `time.json` that lists every recorded color frame’s timestamp on a shared microsecond clock, together with object masks that are named by their canonical timestamp t on the same clock and are shared across cameras. We exploit this annotation to build a single canonical timeline: we first compute the common observation window $[t_{\min}, t_{\max}]$ as the intersection of the four per-camera intervals, then take the union of all canonical mask times that fall inside this window. For each canonical time t_i on this timeline and each camera k , we look up the nearest color frame in k ’s `time.json` and accept it only if it lies within a tolerance of $\pm 16 \text{ ms}$ (approximately half a frame at 30 Hz); otherwise camera k is marked as having no observation at t_i . The tracker then advances the four streams in lockstep over the canonical timeline; a camera with no frame at t_i is simply skipped at that step and ignored as a switching target, which keeps the system robust to the small per-camera drops that occur in practice.

Multi-camera tracking with camera switching. The multi-camera tracker is designed so that a fully occluded view can be replaced by one that still sees the object, while only ever maintaining a single trajectory in a shared world frame. Concretely, the tracker keeps one *active* camera at a time and a single object pose P_t^{world} in the BEHAVE world coordinate system. For each canonical time, P_t^{world} is projected into the active camera using the calibrated Kinect extrinsics, FoundationPose’s `track_one` refines that camera-space pose against the active view’s RGB-D, and the refined pose is promoted back to world coordinates via the same extrinsic. Switching between cameras is driven by the same depth-consistency confidence introduced above, which gives a unified mechanism for both recovery (in single-camera mode) and view selection (in multi-camera mode).

Switching policy. When the confidence on the active view falls below the re-initialization threshold, the current world pose is projected into every other available camera at that time step and conf is recomputed per view. If the best alternative view exceeds the active view by a margin δ (a hysteresis term that prevents oscillation between similarly scored cameras), and at least N_{dwell} frames have elapsed since the last switch, the tracker switches to that camera and re-refines the pose there. The view used to track the object therefore changes as the person moves around it, and

the system gracefully degrades to single-view tracking on canonical times where only one camera has both a valid frame and a clear line of sight.

3.2. GVHMR Human Pose Estimation

We use GVHMR to retrieve per-frame SMPL body parameters from a monocular RGB video. For each BEHAVE sequence, we run GVHMR on the video from the Kinect camera 1 throughout our experiments. GVHMR produces two parallel sets of SMPL parameters per frame: a camera-frame set (`smpl_params_incam`) expressing the body in the primary camera’s coordinate system, and a global-frame set (`smpl_params_global`) expressing it in GVHMR’s gravity-aligned world frame. Each set includes a global orientation in axis-angle form, a body pose, covering 21 articulated joints, a translation, and a shape vector. Together with GVHMR’s estimated per-frame camera intrinsics K , these parameters determine the body geometry at each timestep.

3.3. Depth and Focal-Length Correction

GVHMR assumes a focal length of $f_{\text{GVHMR}} = \sqrt{W^2 + H^2}$, whereas BEHAVE provides calibrated camera intrinsics $f_x \approx 980\text{px}$, which is typically smaller. This discrepancy introduces a systematic depth-scale error in the recovered human trajectory.

To compensate for this effect, we rescale the camera-frame depth component of the SMPL translation using the ratio between the calibrated BEHAVE focal length and the focal length estimated by GVHMR:

$$s = \frac{f_{\text{BEHAVE}}}{f_{\text{GVHMR}}}.$$

The correction is applied before the human enters the joint optimization. This adjustment significantly improves alignment between reconstructed humans and objects.

3.4. SMPL-X to SMPL Alignment

GVHMR predicts body parameters using the SMPL-X model, whereas evaluation within the BEHAVE framework is performed using SMPL. Although the two body models share a compatible body parameterization, differences in their template definitions introduce a systematic offset at the pelvis.

To ensure compatibility between reconstruction and evaluation, we compute a pelvis-origin correction by comparing the pelvis locations of the SMPL-X and SMPL template meshes in their canonical poses. The resulting offset is added to the corrected SMPL translation before mesh generation. This alignment removes a consistent vertical displacement and ensures physically plausible contact between the reconstructed human and object.



Figure 2. **Examples of Human-object interaction reconstruction after joint optimization.** The reconstructed human and object are jointly refined to improve interaction consistency.

3.5. Coordinate Frame Unification

A central challenge in combining GVHMR and FoundationPose is that the two systems operate in different coordinate frames. GVHMR reconstructs human motion in a gravity-aligned global frame, whereas FoundationPose estimates object poses in camera coordinates. Furthermore, BEHAVE defines its own world coordinate system through multi-camera calibration. To enable joint human-object reconstruction, all predictions must be transformed into a common reference frame.

For human reconstruction, we use GVHMR’s camera-frame predictions and transform them into the GVHMR global frame through a world-to-camera transformation estimated from corresponding global and camera-space poses. The resulting transformation is averaged across all frames to obtain a single sequence-level mapping. Human mesh vertices are then transformed into the shared GVHMR global coordinate system.

For object reconstruction, FoundationPose estimates are first transformed into the BEHAVE world frame using the calibrated Kinect extrinsics. These poses are subsequently mapped into the GVHMR kinect-1 camera coordinate system through the world-to-local extrinsic of camera-1. In the multi-camera setting, object poses are already fused into the BEHAVE world frame, allowing the initial camera-to-world conversion to be omitted.

When multiple GVHMR runs are available from different cameras, each reconstruction produces an independent gravity-aligned world frame. To align these frames, we perform body-joint registration using a set of stable skeletal joints and estimate a similarity transformation with the Umeyama algorithm. This step compensates for residual scale and pose inconsistencies between camera-specific reconstructions and produces a unified global coordinate system.

3.6. Joint Optimization

Given the human parameters from GVHMR and the object pose from FoundationPose, the joint optimizer refines

both reconstructions in a unified coordinate frame. For each of the T frames in the sequence, we optimize the SMPL global orientation $\Gamma_t^c \in \mathbb{R}^3$, body pose $\theta_t \in \mathbb{R}^{21 \times 3}$, human translation $\tau_t^c \in \mathbb{R}^3$, object rotation $\Phi_t \in \mathbb{R}^3$ (axis-angle), and object translation $\tau_t^o \in \mathbb{R}^3$. The SMPL body shape coefficients $\beta \in \mathbb{R}^{10}$ and object mesh are kept fixed, since the human’s shape should not change over a sequence and the object geometry provided by BEHAVE is a known input to both FoundationPose and the optimizer.

The objective function combines four terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{data}} + \lambda_2 \mathcal{L}_{\text{contact}} + \lambda_3 \mathcal{L}_{\text{pen}} + \lambda_4 \mathcal{L}_{\text{temp}}$$

Data term: The data term anchors the solution to the initial estimates, ensuring that the optimization does not drift far from the GVHMR and FoundationPose initializations. It is calculated by taking the sum of the MSEs of each parameter and the initial parameter estimates:

$$\begin{aligned} \mathcal{L}_{\text{data}} = & \text{MSE}(\Gamma^c, \Gamma^{c,\text{init}}) + \text{MSE}(\theta, \theta^{\text{init}}) + \\ & \text{MSE}(\tau^c, \tau^{c,\text{init}}) + \text{MSE}(\Phi, \Phi^{\text{init}}) + \\ & \text{MSE}(\tau^o, \tau^{o,\text{init}}). \end{aligned}$$

Contact term: The contact term pulls a set of pre-selected human vertices, corresponding to the body parts that interact with the object in the given sequence, toward the object’s surface. We obtain the set of contact vertex indices $\mathcal{V}_c$ from the standard SMPL body-part segmentation. We selected them based on the interaction in each sequence (e.g. left and right hands for held objects; left and right hands, spine, and shoulders for backpack sequence). At each iteration, we run the SMPL forward pass to compute the posed human vertices, extract the subset of $\mathcal{V}_c$, and compute distances to a fixed set of object surface samples obtained by rotating and translating the surface samples into the current object pose. For each contact vertex we find its nearest object point $\mathbf{o}_{t,\text{NN}(i)}$ (where $\text{NN}(i)$ is the index of the nearest object sample to vertex i), take the top $k = 30$ smallest such distances per frame, and average them:

$$\mathcal{L}_{\text{contact}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{k} \sum_{i \in \text{TopK}_k(d_t)} d_{t,i}, \quad k = 30.$$

where $d_{t,i} = \|\mathbf{v}_{t,i} - \mathbf{o}_{t,\text{NN}(i)}\|_2$. The nearest-neighbour assignment is computed under `torch.no_grad` (`argmin` is not differentiable), and the gradient flows only through the resulting distance to the body and object pose parameters. The top- k restriction ensures that only the vertices closest to the object contribute to contact at each frame, so the loss encourages localized contact rather than dragging the entire body region against the object.

Penetration term: The penetration term penalizes any human vertex that lies inside the object’s surface. We sample N_o object surface points and their outward unit normals once at the beginning of optimization and transform them into the current object pose at each iteration. For a broad set of body vertices $\mathcal{V}_p$ (a random sample unioned with the contact vertices), we find each vertex’s nearest object point and that point’s outward normal $\mathbf{n}_{t,\text{NN}(i)}$ under `torch.no_grad`, then compute the signed distance

$$s_{t,i} = (\mathbf{v}_{t,i} - \mathbf{o}_{t,\text{NN}(i)}) \cdot \mathbf{n}_{t,\text{NN}(i)},$$

which is positive when the vertex lies outside the object and negative inside. The loss is a one-sided ReLU penalty:

$$\mathcal{L}_{\text{pen}} = \frac{1}{|\mathcal{T}| \cdot |\mathcal{V}_p|} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{V}_p} \max(0, -s_{t,i}),$$

where $\mathcal{T} \subset \{1, \dots, T\}$ is a strided subset of frames used for efficiency. This formulation allows the body to touch the object freely while penalizing genuine intersection in proportion to its depth.

Temporal term: The temporal term penalizes frame-to-frame change in both translation and orientation. Translations are differenced directly in Euclidean space, but rotations cannot be differenced in raw axis-angle form since the axis-angle representation is non-injective near $\theta = \pi$, where two numerically opposite vectors encode nearly identical rotations. A Euclidean difference of such axis-angle vectors can produce a discontinuity, which the optimizer attempts to “smooth” by rotating the body through an incorrect orientation. To avoid this, we convert each frame’s rotation parameters to a rotation matrix via the Rodrigues formula and difference the matrices using the chordal distance:

$$\begin{aligned} \mathcal{L}_{\text{temp}} = & \sum_{R \in \mathcal{R}} \frac{1}{T-1} \sum_{t=2}^T \|R_t - R_{t-1}\|_F^2 + \\ & \sum_{\tau \in \{\tau^c, \tau^o\}} \frac{1}{T-1} \sum_{t=2}^T \|\tau_t - \tau_{t-1}\|^2, \end{aligned}$$

where the first sum runs over all rotation matrices (global orientation, body-pose, and the object rotation) and the second over both translations.

Implementation: We implement the SMPL forward pass as a differentiable batched function over all T frames, so that gradients from contact, penetration, and temporal losses propagate back to the pose parameters through the full kinematic chain. The object surface is transformed each iteration by applying the current rotation and translation (Φ_t, τ_t^o) to the static canonical mesh. We optimize all parameters jointly across all frames using Adam with a learning rate of 10^{-4} for 300 iterations. Adam was chosen because the parameter blocks span very different magnitudes,

which suits Adam’s per-parameter adaptive step sizes. After optimization, we apply a final smoothing pass to the predicted trajectories to remove any residual high-frequency variation that survives the temporal loss using the Savitzky-Golay filter along the time axis. Rotations are smoothed by converting to unit quaternions, sign-unrolling consecutive frames so that the sequence remains continuous on a single hemisphere of S^3 , applying the Savitzky-Golay filter to the quaternion components, re-normalizing, and converting back to axis-angle. This avoids the same axis-angle wrap discontinuities that motivated the matrix-space temporal loss. Translations are smoothed directly in Euclidean space.

3.7. Visualization and Rendering

For qualitative evaluation, reconstructed human and object meshes are rendered onto the original BEHAVE color videos. Both meshes are first transformed into the shared GVHMR global coordinate frame and subsequently projected into the target Kinect camera using the calibrated BEHAVE camera parameters. Rendering is performed using PyTorch3D, and the resulting meshes are composited with the original video frames to produce visualization videos. In the multi-camera setting, frame synchronization is achieved using BEHAVE’s hardware timestamps, ensuring temporal consistency across camera views.

4. Experiments

4.1. Setup & Implementation Details

We evaluate on the BEHAVE [1] dataset, which provides synchronised RGB-D from four calibrated Azure Kinects together with ground-truth SMPL-H human fits and rigid object poses for everyday human-object interactions. We focus on a subset of Date02.Sub02 sequences, detailed in Tab. 1, to evaluate our pipeline.

To manage GPU memory constraints (11 GB), we resize the native 2048×1536 video inputs so the shorter side is 480 px, adjusting the color intrinsics accordingly. Depth frames are clipped at a maximum distance of 6 meters. For object tracking initialization and confidence recovery, we utilize the ground-truth object masks provided by the dataset.

4.2. Metrics

Human pose is evaluated against BEHAVE’s ground-truth SMPL-H fits. We report the Procrustes-aligned mean per-joint position error (PA-MPJPE) across the 22 SMPL body joints, and the Procrustes-aligned vertex-to-vertex error (PA-V2V) across the 6890 body vertices. Both metrics are reported in millimeters (lower is better).

Object pose is evaluated against BEHAVE’s per-frame rigid ground-truth poses using 1,000 sampled mesh surface

Table 1. GVHMR per-sequence human pose error on Date02.Sub02. Errors are Procrustes-aligned (in mm), no scale.

Sequence	Frames	PA-MPJPE ↓	PA-V2V ↓
backpack_back	1383	45.3	72.6
boxlarge_hand	1325	47.8	66.6
boxsmall_hand	1423	44.9	64.4
chairblack_hand	1433	44.1	64.3
monitor_hand	1429	84.5	104.4
plasticcontainer	1416	48.8	79.8
stool_move	1419	41.8	60.4
suitcase_ground	1434	49.2	69.5
tablesmall_move	1421	40.2	59.1
Mean	12683	49.6	71.2

points. We report the mean per-vertex L2 distance (ADD), its symmetric variant (ADD-S), geodesic rotation error (degrees), and translation error (millimeters). To account for differing canonical mesh origin conventions between BEHAVE and FoundationPose, we fit a single $SE(3)$ world offset and canonical-frame rigid transform jointly across the sequence. We report the residual error after removing this rigid alignment, referred to as *offset-aligned* ADD, which mirrors standard BOP evaluation practices [2].

4.3. Baselines

We progressively add information to the object tracking pipeline to evaluate each component’s contribution. We compare four configurations, ultimately refining the best-performing object trajectory jointly with the human pose:

- **Tracking only (k1):** The standard FoundationPose single-camera tracker operating on the primary view (k1). Re-initialization is disabled ($\tau=0$). The tracker registers at the first frame and operates purely feed-forward thereafter, serving as our baseline for raw single-view performance.
- **Single-camera pipeline (k1):** Our augmented single-camera baseline on k1, incorporating confidence-based re-initialization ($\tau=0.2$). The tracker automatically re-registers using the per-frame mask whenever depth-tracking confidence falls below the threshold. This isolates the impact of our recovery mechanism independently of multi-view fusion.
- **Multi-camera FoundationPose (ours):** Our proposed multi-view tracking framework that maintains a single, unified pose within the BEHAVE world frame. It actively switches the reference camera during confidence drops (using hysteresis) and smoothly interpolates poses across occlusion gaps where no valid mask is observed ($\tau=0.2$).

- **Multi-camera FP + joint optimization (ours):** The full proposed pipeline. The multi-camera object pose is jointly optimized alongside the GVHMR-recovered human mesh, enforcing contact, penetration, and temporal smoothness constraints to ensure physically plausible interactions.

4.4. Results

Human pose. GVHMR provides stable human motion recovery (Tab. 1), averaging 49.6 mm PA-MPJPE and 71.2 mm PA-V2V. The only major outlier is `monitor_hand` (84.5 mm PA-MPJPE), where severe torso occlusion causes the characteristic monocular body-collapse failure mode.

Object pose. Table 2 reports object pose errors. Overall, our multi-camera tracker drastically reduces mean ADD to 23.6 mm (a $7.8\times$ improvement over tracking-only) and rotation error to 4.46° . However, this average is heavily skewed by occlusion-heavy sequences. Excluding `backpack_back` and `stool_move`, baseline single-camera tracking is already competitive (15.3 mm vs 22.9 mm ADD), indicating our framework’s primary advantage is robustness to occlusion rather than baseline accuracy.

For instance, in `stool_move`, the `k1` view loses the object twice causing permanent drift in the tracking-only baseline (ADD 346.7 mm). Even single-camera re-initialization cannot fully recover (119.0 mm). By dynamically handing off tracking to unoccluded views, our multi-camera pipeline maintains a stable lock, yielding an ADD of 52.9 mm.

Joint optimization. The joint optimisation step refines the multi-camera object trajectory and the GVHMR-recovered human under the contact, penetration and temporal losses of Sec. 3.6. To assess the resulting reconstruction end-to-end we report the joint Procrustes-aligned vertex-to-vertex error (PA-V2V): a single rigid transform is fit per frame to the concatenated human and object vertices, so a residual reflects inconsistency in their *relative* pose rather than absolute placement. Across the seven optimised sequences (9,848 frames), PA-V2V averages 93.0 mm for the human and 226.2 mm for the object. The remaining object error is dominated by the body-occluded sequences `chairblack_hand` (677 mm) and `tablesmall_move` (235 mm); on cleanly visible interactions it stays in the 113–144 mm range. This is consistent with the multi-camera object trajectory being retained largely intact under the data-term anchor while the contact and penetration losses tighten the human–object relative pose.

4.5. Failure cases

Geometrically ambiguous objects. Single-camera tracking struggles with locally symmetric objects, as demonstrated in the `backpack_back` sequence. From the `k1` viewpoint, the backpack’s front and back are nearly identical. Once the participant turns, single-camera baselines cannot distinguish a 0° from a 180° rotation and drift into a permanently flipped state (ADD > 800 mm, rotation $\approx 90^\circ$ – 100°). Crucially, the single-view depth-tracking confidence remains high because the flipped pose plausibly explains the observed RGB-D data. This false confidence even causes the tracker to mistakenly latch onto the participant’s leg after brief occlusions, as the leg shares a locally similar surface geometry. Since single-camera re-initialization relies on this flawed confidence signal, it fails to recover the track. In contrast, our multi-camera tracker resolves this ambiguity: conflicting multi-view observations lower the overall confidence, triggering a camera hand-off that recovers the true pose (35.9 mm ADD, 6.17° rotation).

Severe self-occlusion in monocular human pose. Monocular human mesh recovery is highly vulnerable to prolonged occlusion. In the `monitor_hand` sequence, the participant holds a monitor tightly against their chest. Deprived of torso observations, GVHMR’s depth prior is dominated by the head and upper arms, causing the predicted SMPL mesh to collapse inward (84.5 mm PA-MPJPE, $\approx 1.7\times$ the sequence average). Notably, the object pose estimation on this same sequence is nearly perfect (5.0 mm ADD, 1.48° rotation), as the monitor’s planar, highly textured surface is ideal for FoundationPose. This highlights a distinct challenge in HOI reconstruction: severe occlusions can catastrophically degrade the human mesh recovery while leaving object tracking unaffected.

5. Conclusion

We presented a hybrid pipeline for reconstructing 3D human-object interactions by integrating monocular human motion recovery (GVHMR) with multi-view object pose estimation (FoundationPose). To unify these independent systems, we introduced coordinate-frame alignment, depth-scale correction, and a SMPL-X-to-SMPL pelvis offset. The resulting trajectories are jointly optimized using contact, penetration, and temporal smoothness constraints to ensure physical plausibility.

Evaluations on the BEHAVE dataset demonstrate the effectiveness of this hybrid approach, while also confirming that occlusion remains the primary bottleneck. Interactions involving body-worn or heavily occluded objects are substantially more challenging to reconstruct than simple hand-held manipulation, indicating that continuous visibility is critical.

Table 2. Object pose error on Date02_Sub02. Offset-aligned ADD and ADD-S are in mm, rotation in degrees, translation in mm. Lower is better. “Tracking only” is the baseline FoundationPose without re-initialization; “Single-cam pipeline” includes our confidence-based re-init on k1; “Multi-cam” is our shared-world tracker.

Sequence	Tracking only (k1)				Single-cam pipeline (k1)				Multi-cam (ours)			
	ADD	ADD-S	rot	t	ADD	ADD-S	rot	t	ADD	ADD-S	rot	t
backpack_back	824.0	567.5	90.20	1094.8	923.4	669.7	102.52	1142.4	35.9	20.3	6.17	78.7
boxlarge_hand	34.7	22.5	3.61	31.8	22.3	17.0	2.41	20.3	20.9	16.3	2.18	19.3
chairblack_hand	26.1	17.9	2.83	40.9	18.6	14.3	2.18	33.7	20.1	15.2	1.91	30.9
monitor_hand	4.6	4.1	1.32	3.2	4.8	4.2	1.34	3.4	5.0	4.7	1.48	3.3
stool_move	346.7	193.1	34.48	330.6	119.0	71.9	12.40	115.1	52.9	20.6	15.18	31.3
suitcase_ground	22.4	15.6	3.20	20.2	16.8	12.9	2.65	14.8	13.9	11.1	2.34	12.0
tablesmall_move	26.6	14.8	2.70	55.0	23.4	13.8	2.46	47.2	16.4	11.2	1.93	38.7
Mean	183.6	119.4	19.76	225.2	161.2	114.8	17.99	196.7	23.6	14.2	4.46	30.6

Table 3. Joint optimisation: joint Procrustes-aligned V2V error (mm). Lower is better.

Sequence	Frames	Human ↓	Object ↓
backpack_back	1381	99.0	134.4
boxlarge_hand	1323	70.4	113.1
chairblack_hand	1433	151.0	677.4
monitor_hand	1427	102.4	137.3
stool_move	1430	68.7	143.6
suitcase_ground	1432	77.9	129.8
tablesmall_move	1422	79.7	235.1
Mean (n=7)	9848	93.0	226.2

The main limitation of our framework is its reliance on independent upstream predictions; severe initial errors in body shape or object tracking cannot always be recovered during post-processing optimization. Future work should address this through end-to-end joint estimation models, interaction-aware priors, and by extending multi-view or depth-aware processing directly to the human reconstruction component to improve overall geometric robustness.

6. Contributions of team members

The work was divided according to the main components of the project. Nina Gruteser focused on the implementation of GVHMR, the joint optimization framework, and the associated loss functions. Mathilda Lee was responsible for coordinate-frame unification, including depth and focal-length correction, coordinate alignment, and visualization of the reconstructed results. Marco Weder implemented the FoundationPose pipeline, multi-camera object reconstruction, and evaluation. All team members contributed to the integration of the system, analysis of results, and preparation of the report and presentation.

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